Bone Stress Injuries of the Foot

Background

- Bone stress injuries (BSI), commonly known as "stress fractures," can range in severity. BSI usually result from repeated stress to a bone over time from activities such as running, jumping, or marching.
- Women and people with low physical fitness, high arches, low bone density, or who've had previous injuries or BSI have a higher risk of developing BSI.
- Additional risk factors include rapid increases in training intensity or volume, inadequate recovery time, poor running form, a poor or inadequate diet (insufficient calories, vitamin D, iron, etc.), smoking, and, for women, irregular menstrual periods.
- Primary symptoms of BSI are localized pain and tenderness. BSI often hurts at night or at rest, and pain can worsen with continued high-impact activities.
- Diagnosis is made through medical imaging. X-ray is the first imaging test used, but in some cases an MRI or a bone scan is required for diagnosis or further evaluation.
- Treatment includes rest, activity modification, and rehabilitation. Depending on the location and severity, some BSI may require surgery. Most foot BSI heal well in about 4–6 weeks, but returning to sports may take longer.

Latest Research

- Inadequate nutrition, especially a lack of total calories, in both male and female athletes can cause physiological stress and suppress important hormones, leading to weakened bones. Your doctor may order lab and imaging tests to assess your nutrition and bone health.



Rehab Instructions

Phase 1: On average, this phase lasts 3–14 days.

- Rest your injured foot using the method prescribed by your doctor. This can include crutches, casting, or a walking boot.
- Control pain with acetaminophen (Tylenol) and apply ice to the area for 20 minutes at a time, up to 2–3 times daily. Avoid NSAIDs such as ibuprofen (Aleve, Advil, Motrin, etc.).

Phase 2: On average, this phase lasts 4–6 weeks.

- Begin weight-bearing with or without crutches as advised by your doctor.
- Start low-impact cross-training activities such as riding a recumbent bike, swimming, deep water running*, or using an antigravity treadmill.
- Avoid impact and high-impact activities such as running or jumping.
- Control pain with acetaminophen, ice, or NSAIDs as needed.

Phase 3: Begin this phase after 2–3 weeks of pain-free walking and low-impact activities.

- Resume running at a slow pace and minimal distance every other day. Gradually increase the distance and duration of exercise over 4 weeks. Then, increase your pace and intensity.
- Start resistance or weight training 2–3 times per week and focus on strengthening your legs and core, flexibility, and balance.
- Optimize your running form by getting a gait analysis and use physical therapy to improve your strength and balance. This will reduce the chance of a BSI recurrence.

Practical Tips For Success

- Incorporate cross-training and lower body muscle endurance training (high repetitions, low resistance) 2–3 times per week.
- **Follow the 10% rule**. In general, do not increase your activity level (for example, your running mileage) by more than 10% from week to week.
- **Optimize your diet.** Make sure you're consuming adequate amounts of protein, vitamin D, calcium, and fresh fruits and vegetables. See a registered dietitian (RD) for help with dialing in your diet.
- Listen to your body. Rest when needed and avoid over-training.



*Deep Water Running (DWR)

- DWR is a form of cross-training that you can do once you've progressed to Phase 2 of your injury rehab.
- Submerge into water deep enough so your shoulders are just under water when your body is in a slightly forward position.
- You can use a flotation belt or tuck a pool noodle into the back of a regular belt. Or, you can put the pool noodle under your arms to help you float.
- Your arm motion should be the same as when you run on land. Don't make contact with the bottom of the pool.

References

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